CS_490_RND

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1 Project Summary

This project collect user feedback after cellular call and correlate it with network parameters. The Android system automatically collect data after a phone call ends and send it to a server (in the project's case a python server) to store and peform analysis.

2 System Overview

The system follows a simple client-server architecture.

Android Client: Detects call end events, shows a feedback pop up, collects user input and network data.

Python Server: Receives feedback data and stores it in a file.

Communication between the Android app and server happens using HTTP POST requests with JSON data using a REST API.

3 Android Components and Their Function

3.1 Broadcast Receiver

- Listens for system events such as **PHONE_STATE** and **BOOT_COMPLETED**.
- Detects when a call ends (IDLE state).
- Starts the pop up service only when required.

Purpose: Saving resources by enabling even driven execution rather then continuously running background service.

3.2 Call State Logic

- Ignores RINGING and OFFHOOK states.
- Triggers action only when phone state becomes IDLE.
- Ensures feedback is shown only after the call ends.

3.3 PopUP Service

- Displays a floating feedback form using **WindowManager**.
- Collects user ratings such as call quality and audio issues.
- Reads system data like signal strength, network type, and location.
- Sends collected data to the backend server.
- Removes itself after submission to avoid resource leakage.

Purpose: Collects feedback without interrupting user workflow.

4 Data Collected

4.1 User-Reported Data

- Overall call quality (Good ,Fair, Poor)
- Audio issues (echo, noise, call drop, etc.)
- Environment (indoor, outdoor, vehicle)
- Optional user comments

4.2 Device-Reported Data

- Network type (2G , 3G , 4G , 5G)
- Signal strength in dBm
- GPS location (latitude, longitude)
- Timestamp of call end

5 Backend Python Server

5.1 Server Functionality

- Implemented using Python Flask.
- Exposes a REST API endpoint feedback.
- Accepts JSON data via HTTP POST.
- Appends received data to a **.jsonl** file.

5.2 Data Storage

- Uses JSON Lines format for simplicity.
- Each line represents one feedback entry.
- Fast write performance and low overhead.

6 Testing and Results

- Tested on multiple Android devices and versions.
- Pop up appeared successfully after every call end.
- Signal strength readings matched system values closely.
- Data transmission to server was reliable.

7 Conclusion

The project uses automated method to collect objective quality of real world cellular call and subjective QOE experience via pop up(overlay) and combining it to provide more valuable ,robust network measurements.